

Background and 511 keV line sensitivity of a balloon-borne transition-edge sensor Laue-lens telescope

Authors omitted for review^a

^a*Affiliations omitted for review*

Abstract

The Galactic 511 keV annihilation line is a direct tracer of low-energy positrons, but its compact-source content and narrow-line kinematics remain difficult to measure because MeV observations are strongly background dominated. We evaluate a balloon-borne pointed 511 keV telescope that combines a Ge(111) Laue lens with a transition-edge-sensor (TES) microcalorimeter focal plane. The detector-coupled Monte Carlo model transports the focused optical signal, prompt atmospheric backgrounds, and delayed activation through the same engineering detector/cryostat mass model. The same event processing then applies the active-shield veto, Compton/field-of-view consistency selection, narrow line window, and mission-time rate fold. The baseline optical configuration has a 10 m focal length, $A_{\text{eff}}(511 \text{ keV}) = 20.08 \text{ cm}^2$, and a 510.58–511.42 keV analysis window; delayed activity is emitted from sampled production positions recorded during activation transport. In the current fix5 full-statistics closure, a reference point-source flux of $10^{-4} \text{ ph cm}^{-2} \text{ s}^{-1}$ gives day-15 selected line-window background and signal rates of $3.92 \times 10^{-2} \text{ s}^{-1}$ and $1.19 \times 10^{-3} \text{ s}^{-1}$, respectively. Folding these selected rates over the mission-time profile gives $Z_{20\text{d}} = 7.80$, a 3σ detection time of 2.51 d, and a 20-d 3σ flux threshold of $3.85 \times 10^{-5} \text{ cm}^{-2} \text{ s}^{-1}$. The fixed day-15 delayed inventory is 85.449 Bq after the NUBASE ground-state correction, and the selected W/collimator-origin delayed audit finds zero W/collimator-origin selected events. Relative to the previous v3p5 exact-position authority, the current closure reduces the selected hard-window background to 0.623 of the reference value while preserving the focused signal response at 1.004. Secondary spatial-analysis and atmospheric-line-of-sight checks remain diagnostics; the hard-window fix5 calculation is the primary rate result. These results define a detector-coupled selected-rate basis for evaluating focused TES

spectroscopy in narrow-line point-source searches, provided activation and prompt atmospheric backgrounds are modeled in the same selection space as the signal.

Keywords: Gamma-ray instrumentation, 511 keV annihilation line, transition-edge sensors, Laue lens, balloon-borne telescope, Monte Carlo background simulation, activation background

1. Introduction

The 511 keV gamma-ray line from electron–positron annihilation is a direct spectroscopic tracer of low-energy antimatter in the Galaxy. After positrons slow down in the interstellar medium, they annihilate either directly with electrons, producing two photons near 511 keV, or through positronium formation, producing the 511 keV line together with a three-photon continuum below the line. This radiative complex probes positron production, propagation, thermalization, and annihilation conditions [1]. The line was first reported from the Galactic-center direction by balloon-borne experiments and later identified unambiguously with high-resolution Ge detectors as positron-annihilation radiation [1]. The observed Galactic flux implies a positron annihilation rate of order $10^{43} \text{ e}^+ \text{ s}^{-1}$, making the origin of these positrons a long-standing problem in MeV gamma-ray astronomy [1, 4].

The unresolved source population cannot be inferred from source inventories alone. Candidate channels include β^+ decay of radioisotopes from stellar explosions and novae, pair creation in compact objects, cosmic-ray interactions, and possible non-standard particle channels [1–3]. The observed annihilation morphology also depends on positron propagation before thermalization and annihilation. Spectroscopy constrains the annihilation medium through the line centroid, line width, and positronium fraction, while continuum limits above 511 keV constrain the injection energy [1, 2]. The observables that would most directly separate source scenarios, including compact-source content, small-scale structure, velocity shifts, and spatially resolved line broadening, remain limited by instrumental sensitivity, angular resolution, and background systematics.

The current observational picture has been shaped by OSSE/CGRO, INTEGRAL/SPI, and COSI. INTEGRAL/SPI established a bright bulge-like component and fainter disk emission, with the detailed bulge/disk decomposition depending on the adopted sky model and instrumental-background

treatment [2, 3]. Recent long-exposure SPI analyses recover the central, broad-bulge, and disk structures and suggest marginal additional structures associated with massive-star regions [3]. COSI provided an independent balloon measurement from a 46-d super-pressure-balloon flight, detecting the Galactic 511 keV signal with Compton imaging and finding that the emission is broader than a single point source [4, 5]. Together, these results show that MeV line measurements are limited by weak celestial signals and strong, time-dependent atmospheric and instrumental backgrounds.

Wide-field Compton telescopes are essential for mapping the large-scale 511 keV sky. A complementary strategy is to use a pointed telescope that concentrates 511 keV photons onto a small, high-resolution focal plane. Such a system tests a narrower question: whether a known direction contains compact line emission from an unresolved source, a bright central core, or a transient pair-plasma episode. The annihilation signatures reported during the 2015 outburst of V404 Cygni illustrate why targeted compact-object follow-up is relevant to the broader positron problem [6]. A focused payload therefore does not replace wide-field Compton missions; it addresses a different observable in a smaller field of view.

The 511-CAM concept follows this route by combining gamma-ray concentrating optics with stacked transition-edge-sensor microcalorimeter arrays [7]. Its relevant capability is the combination of photon concentration and sub-keV calorimetric energy resolution, two properties rarely available together at MeV energies. A projected 511 keV detector-assembly resolution of a few 10^2 eV FWHM corresponds to a velocity scale of a few 10^2 km s $^{-1}$ and could enable velocity-shift or line-broadening measurements in favorable cases [7]. Event topology also enters the selection. Single-pixel photoelectric events provide direct calorimetric spectroscopy, while multi-pixel events can be checked against Compton kinematics and rejected if inconsistent with photons arriving through the focusing optics.

The main technical limit is background. Balloon-borne gamma-ray spectrometers operate in a strong radiation environment and are necessarily background dominated. Classical studies of balloon-borne Ge spectrometers identify atmospheric-neutron scattering, atmospheric and cosmic gamma-ray aperture flux, beta decays of activation products, and shield leakage as major background components; they also emphasize minimizing passive material near the detector and lowering effective shield thresholds where possible [14]. For a 511 keV line payload, this limit is more direct because the science signal lies at the same energy as strong atmospheric and instrumental annihilation

features. A credible sensitivity estimate therefore requires a detector-coupled model of the optics, entrance windows, cryostat, passive material, active shield, prompt atmospheric components, and delayed activation.

Detailed Monte Carlo mass modeling is the standard route to such estimates. The INTEGRAL/SPI response production required a high-fidelity mass model, ground- and in-flight-calibration comparisons, and a practical separation between imaging response and redistribution response to control computing and storage costs [17]. Modern MeV mission studies, including AMEGO and COSI, similarly rely on Geant4/MEGAlib-based simulations to calculate response, background, event reconstruction, and sensitivity [18–22]. The same philosophy is adopted here, but applied to a narrower and more demanding target: a balloon-borne, focused, narrow-line 511 keV telescope with a TES focal plane. Because the study is simulation based, the transport code is not treated as evidence by itself. The rate result is tied to explicit geometry and source normalizations, tracked focal-plane crossings, recorded activation-production positions, common detector-level selections, and downstream convergence or boundary checks.

We evaluate that payload at the selected-rate level. The simulation chain starts from a Ge(111) Laue-lens source model at 511 keV and transports focused photons through the detector entrance window into an engineering detector/cryostat mass model. The modeled assembly includes the TES focal plane, staged thermal windows, a Be entrance window, Cu/W passive shielding, cryogenic service proxies, and an active CsI shield. The background chain includes prompt atmospheric components, activation buildup, exact-position delayed-source transport, active-shield vetoes, Compton/field-of-view consistency selections, a narrow 511 keV line window, mission-time rate evolution, and counting-significance estimates. The output is a closed rate calculation in which signal and background pass through the same detector-coupled selection definitions, rather than an isolated effective-area estimate.

The remainder of the paper is organized as follows. Section 2 summarizes the science-driven instrument requirements. Section 3 describes the simulation chain. Sections 4–7 describe the detector geometry, optics bridge, background model, and event selections. Section 9 presents the sensitivity, convergence, and background-composition results. Section 10 discusses interpretation, limitations, and remaining validation steps. Section 11 summarizes the main conclusions.

2. Science and instrument requirements

The instrument considered here is optimized for a pointed, narrow-line measurement rather than all-sky imaging. The primary science case is a long-duration balloon search for compact 511 keV emission at fluxes of order $10^{-4} \text{ cm}^{-2} \text{ s}^{-1}$ or below. This flux scale is used here as a design target for bright central-core, compact-source, and transient pair-plasma scenarios while remaining complementary to wide-field Compton surveys.

The requirements are stated at the level needed for the present simulation, not as final flight-system specifications. They define what the model must preserve for a credible selected-rate estimate: the source flux normalization, the focused entrance phase space, the narrow line-window response, and the same detector selection for signal and background. This distinction is retained because several quantities, including the TES energy response and final cryostat layout, remain design assumptions or engineering proxies.

Four requirements follow. First, the telescope must concentrate 511 keV photons so that a small focal plane can be used without sacrificing collecting area. Second, the focal-plane response must support a sub-keV analysis window so that the line-window background scales with a narrow spectral interval rather than a broad MeV-band continuum. Third, the detector and shield model must suppress atmospheric, activation, and side-entry backgrounds while explicitly accounting for passive material near the TES absorbers. Fourth, the simulation chain must treat prompt background, delayed activation, focused signal, vetoes, event topology, and mission-time variation consistently. Table 1 summarizes how these requirements map onto the current fix5 exact-position implementation.

These requirements also set the interpretation of the results. A hard-window sensitivity is the primary result because it depends only on energy selection, vetoing, topology/FoV consistency, and the mission-time rate fold. Focused-spot and annular secondary analyses test spatial-analysis boundaries, but they do not replace the hard-window requirement until a final spatial-spectral likelihood and background-nuisance treatment are in place.

3. Simulation workflow

Figure 1 summarizes the workflow. The calculation is divided into geometry definition, prompt and buildup transport, delayed-source construction, focused-optics transport, detector-coupled signal replay, detector

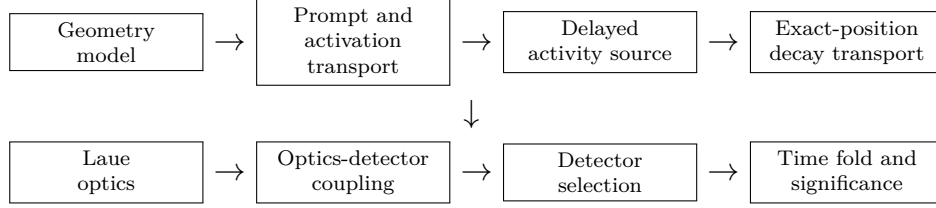


Figure 1: Simulation chain used in this work. The focused signal and all background streams are evaluated in the same detector-response and event-selection space before mission-time significance is calculated.

event selection, mission-time rate folding, source-case normalization, and cumulative significance evaluation. The present manuscript uses the current fix5 position-preserving delayed-source chain as the primary rate calculation. This chain is called “exact-position” below: the delayed decays are sampled from the recorded radioactive-production positions instead of being redistributed into an axisymmetric radial profile. The radial-profile delayed-source chain is retained only as a cross-check.

The primary rate calculation is the current fix5 exact-position delayed-source chain. It uses full-statistics prompt and activation-buildup transport, a fixed day-15 radionuclide population, sampled equal-activity point sources at recorded production positions, and full downstream propagation through the detector response, event selection, and significance calculation. The older homogeneous-beam and radial-profile delayed-source chains are retained only as cross-checks and are not used for the primary sensitivities.

Table 2 gives the traceable version of this chain. Each layer has a defined input, a physical or statistical operation, an output consumed by later layers, and an interpretation boundary. Geometry definition fixes the mass model and coordinate policy but does not produce a rate. Prompt transport and activation transport define direct atmospheric backgrounds and isotope-production histories. The delayed-source layer converts those histories into a fixed day-15 radioactive population and transports sampled exact-position decays. The optical Monte Carlo is signal-only: it supplies the Laue-lens effective area and tracked focused phase space, while the optics-detector coupling and detector-response layers convert that phase space into a detector-coupled point-source response. Mission-time folding and cumulative counting then convert selected rates into sensitivity.

The numerical hierarchy used below follows the same logic. Day-15 selected rates are quoted after the focused signal stream has been scaled by

the 10 m Ge(111) effective area and the reference atmospheric transmission. Mission-mean rates are quoted from the time-dependent fold, and the primary Z , $T_{3\sigma}$, and flux thresholds are quoted from the cumulative significance calculation. Exact-position M-sampling stability is quoted only after the sampled source has been transported through the same detector selection and significance chain. This separation avoids mixing unit-injection rates, day-15 direct expectations, and mission-folded sensitivities in a single result.

Table 1: Traceability from science drivers to instrument and simulation requirements. The table summarizes requirements used to define the current fix5 exact-position analysis; it is not a separate sensitivity calculation.

Science driver	Requirement	Current implementation or evidence	Workflow component
Compact 511 keV targets at $\sim 10^{-4} \text{ cm}^{-2} \text{ s}^{-1}$ or below	Concentrate 511 keV photons onto a small focal plane	10 m Ge(111) Laue lens with $A_{\text{eff}}(511 \text{ keV}) = 20.08476 \text{ cm}^2$; tracked focal-plane photons passing the Be entrance aperture are replayed through the detector model	Laue optics; detector coupling
Narrow-line spectroscopy rather than broad-band counting	Use a sub-keV analysis window and preserve source-flux normalization	$W_{511} = 510.58\text{--}511.42 \text{ keV}$; selected signal $0.00118587 \text{ s}^{-1}$ at $F_0 = 10^{-4} \text{ cm}^{-2} \text{ s}^{-1}$; source-flux response $11.8587 \text{ cps}/(\text{ph cm}^{-2} \text{ s}^{-1})$	Detector selection; source normalization
Balloon prompt, side-entry, and activation backgrounds	Apply detector-coupled veto and topology selections to signal and background streams	CsI post-processing veto, side-entry Compton/FoV consistency, exact-position delayed transport; selected W_{511} background 0.0392162 s^{-1} split into prompt 0.0366410 s^{-1} and delayed $0.00257520 \text{ s}^{-1}$; selected W/collimator-origin delayed audit gives zero selected W/collimator events	Common detector selection; delayed activation audit
Mission-level sensitivity result	Fold rates through mission time and test delayed-source sampling stability	Final cumulative calculation gives $Z_{20\text{d}} = 7.79950$ and $F_{3\sigma}(20\text{d}) = 3.84640 \times 10^{-5} \text{ cm}^{-2} \text{ s}^{-1}$; relative to the previous v3p5 authority, fix5 reduces total selected background to 0.623 and preserves the signal response at 1.004	Mission-time fold; promotion checks

Table 2: Layered simulation workflow used for the current fix5 exact-position rate calculation. The table is a manuscript-facing compression of the validated workflow; detailed validation files retain the run-level details.

Layer	Role in the calculation	Current implementation or evidence	Boundary for interpretation
Geometry model	Define detector mass model, active volumes, side-entry frame, and injection surfaces	Current fix5 detector/cryostat geometry: TES array, Be side window, revised Cu/W passive material and collimator, segmented CsI shield; window tilted 45° upward	Geometry closure only; not a rate calculation
Prompt and activation transport	Generate prompt atmospheric events and activation histories	Detector/cryostat prompt and buildup streams, each with 25,210,216 particles; prompt rates use per-species exposure normalization	Low-statistics checks are closure tests; upstream optics mass is outside the primary rate budget
Delayed activity source	Convert activation products to a fixed day-15 delayed source	Detector/cryostat delayed activity 85.449203 Bq after NUBASE ground-state correction; exact-position sampled point sources with $M = 50000$, seed 260613, and 10^6 transported delayed decays	Activity preserving; source-inventory and W/collimator audits test delayed normalization; upstream Ge-proxy delayed activity is a separate side component
Laue optics	Compute 511 keV effective area and focused focal-plane phase space	Ge(111), 10 m optical Monte Carlo from XOP/CRYSTAL rocking curves; $A_{\text{eff}}(511) = 20.08476 \text{ cm}^2$ and spot contained by Be aperture	Signal only; focal crossings are transported 511 keV gammas, not optics self-background
Optics-detector coupling	Replay tracked optical focal crossings in the detector	Accepted focal gamma rays are converted to a detector transport source and transported through the current fix5 setup	Tracked focal crossings only; signal replay does not represent optics self-background
Aux. optics hardware	Check upstream-lens prompt/activation boundary	Equal-volume active-Ge lens proxy at 10 m; far-field source radius 1060 cm from a 1008.8 cm detector-plus-lens bound; eight EXPACS/PARMA families close prompt and activation-production transport; isolated Ge-proxy delayed transport selects zero W_{511} events, with a 95% upper rate $1.25 \times 10^{-4} \text{ s}^{-1}$	Delayed Ge-proxy side component closed; prompt self-background isolation and support hardware remain outside the primary budget
Detector response	Apply common selection to prompt, delayed, and focused streams	Active-shield post-processing veto, side-entry Compton/FoV consistency, and W_{511} hard window; day-15 W_{511} rates $B/S = 0.0392162/0.00118587 \text{ s}^{-1}$	Veto defined by post-processing selection, not production-time trigger filtering
Mission-time fold	Propagate selected rates through balloon mission time	Time-dependent prompt, delayed activity, atmosphere, and accidental-veto live-factor fold; mean W_{511} rates $B/S = 0.0393546/0.00117748 \text{ s}^{-1}$	Rate-level fold; no new transport at each time bin
Source normalization	Map detector response to source-flux units	W_{511} response $11.8587 \text{ cps}/(\text{ph cm}^{-2} \text{ s}^{-1})$ for the reference point-source case	Point-source normalization kept separate from diffuse-sky checks
Counting significance	Compute cumulative significance and flux thresholds	Hard-window result: $Z_{20\text{d}} = 7.79950$, $T_{3\sigma} = 2.50573 \text{ d}$, $F_{3\sigma}(20\text{d}) = 3.84640 \times 10^{-5} \text{ cm}^{-2} \text{ s}^{-1}$	Gaussian counting metric for consistent comparisons
Robustness checks	Test delayed-source normalization and promotion at selected-rate level	fix5 promotion decision passes against the previous v3p5 authority: background ratio 0.6227, delayed ratio 0.6607, signal-keep ratio 1.0040, and W/collimator selected delayed rate 0	Supports fix5 exact-position delayed transport for the primary rates

4. Detector and cryostat mass model

The detector model is an engineering Geant4/MEGAlib proxy mass model for a balloon-borne TES focal-plane assembly mounted at the cold end of a compact dilution-refrigerator (DR) stack [7, 12]. It is not a final flight CAD model, and it is not intended to reproduce the microfabricated TES layers, detailed wiring, readout layout, or every cryogenic service component. Instead, the model preserves the material columns and local apertures that are first-order for 511 keV transmission, scattering, activation, active-shield vetoing, and event topology. The baseline detector/cryostat geometry therefore contains the TES absorber stack, substrate and window proxies, the side Be entrance window, passive shield/collimator material, cryogenic service-mass proxies, and a segmented CsI active anticoincidence shield. Figure 2 shows the two-dimensional geometry views used as the manuscript-facing mass-model schematic, and Table 3 lists the detector and analysis parameters fixed before the rate calculation.

The DR and service hardware are represented at proxy-mass level. The model keeps the physical order of the main temperature stages: the TES module is placed at the 50 mK/MXC end, followed by cold-plate and heat-exchanger proxies for the CP, still, 4 K, 60 K, and 300 K service regions. Al shells and windows represent staged radiation shields and the vacuum jacket. Support structures, capillaries, cable bundles, readout boxes, and other services are not laid out as flight-ready parts; they are included as local service-mass proxies where they can contribute scattering or activation. This simplification keeps the detector bay, beam aperture, and upper service tower in their correct spatial relationship without treating the model as a mechanical design.

The TES focal plane is represented by the gamma-ray absorber array plus a parameterized calorimeter response, rather than by a full microfabricated TES/readout layout [13]. In the present detector concept the TES thermometer film is an AlMn layer of order 200 nm thickness, whereas the gamma-ray absorber is a millimetre-scale metal structure. For the non-thermal detector-background simulation in this paper, 511 keV stopping, Compton scattering, and activation are therefore governed mainly by the absorber and nearby support mass. The explicit sensitive volume is six layers of 376 Ta absorber pixels, or 2256 pixels in total. Each pixel is a $1.5 \times 1.5 \times 3.0 \text{ mm}^3$ absorber in the side-entry coordinate system, with the 3.0 mm depth aligned to the focused beam. TES-film, wiring, and readout details enter through the assumed

energy response and through service-mass proxies rather than as explicit interaction volumes. The detector map uses a Gaussian TES energy-response proxy with $\sigma = 0.14 \text{ keV}$ for the Ta calorimeter channels; the hard analysis window is then fixed separately as $W_{511} = 510.58\text{--}511.42 \text{ keV}$. The detector efficiency is not imposed as a standalone scalar in this mass-model section; it is generated by Geant4 transport through the absorber/shield geometry and by the downstream event-selection and signal-normalization chain.

The local side-entry geometry points the Be window 45° upward in the zenith-frame source coordinates, with an instrument-frame rotation of 0, 45, 0 degrees. The side aperture passes through the CsI shield and passive package and is bounded by a 2 cm-thick W square-bore sleeve. This sleeve is an aperture and leakage-suppression proxy, not a pixel-matched 376-hole collimator. The outer package uses an Al mechanical shell, Cu/W graded passive liners in the detector bay, a W bottom plate, and segmented side, bottom, and top CsI volumes. Low-Z Al foils and a $150 \mu\text{m}$ Be vacuum window are placed on the side beam line to retain a transparent 511 keV entrance path while keeping cryogenic-window material in the transport model.

The CsI scintillator volumes are used as the mass-model proxy for the active anticoincidence layer. They enter the simulation as materials for transport and activation, and as shield volumes whose deposited energy is evaluated in post-processing. Candidate events are grouped on a common Poisson time axis and vetoed if the summed CsI shield energy in the coincidence window exceeds the analysis threshold. For the baseline CsI configuration the analysis veto threshold is 50 keV and the coincidence window is $1 \mu\text{s}$. This keeps the veto definition common to prompt, delayed, and focused-science streams and prevents production-time trigger flags from becoming the quantitative rate definition. The geometry-closure stage fixes the material and coordinate choices before rate calculations are made; it does not by itself define a sensitivity.

Table 3: Key detector and analysis parameters used for the primary exact-position chain.

Parameter	Value or policy
Detector concept	TES microcalorimeter focal plane
TES pixel copies	$6 \times 376 = 2256$
TES absorber proxy	Ta pixels, $1.5 \times 1.5 \times 3.0 \text{ mm}^3$ each
TES energy-response proxy	Gaussian $\sigma = 0.14 \text{ keV}$ in the detector map
Detector efficiency	Geant4 transport plus downstream selected-rate normalization; no standalone scalar efficiency is imposed here
Active shield	Segmented CsI, post-processing veto
CsI veto threshold	50 keV summed shield energy
Entrance-window stack	staged Al foils plus $150 \mu\text{m}$ Be side window
Passive shield-ing/collimation	Cu/W proxy liners, W bottom plate, and 2 cm W side sleeve
DR/service proxies	MXC/CP/still/4 K/60 K/300 K stage order plus local structural, cabling, and read-out service-mass proxies
Pointing	Local side window tilted 45° upward
Detector trigger/veto definition	Common detector-level post-processing selection
Coincidence window	$1 \mu\text{s}$
Line window W_{511}	510.58–511.42 keV
Reference point-source flux F_0	$10^{-4} \text{ cm}^{-2} \text{ s}^{-1}$

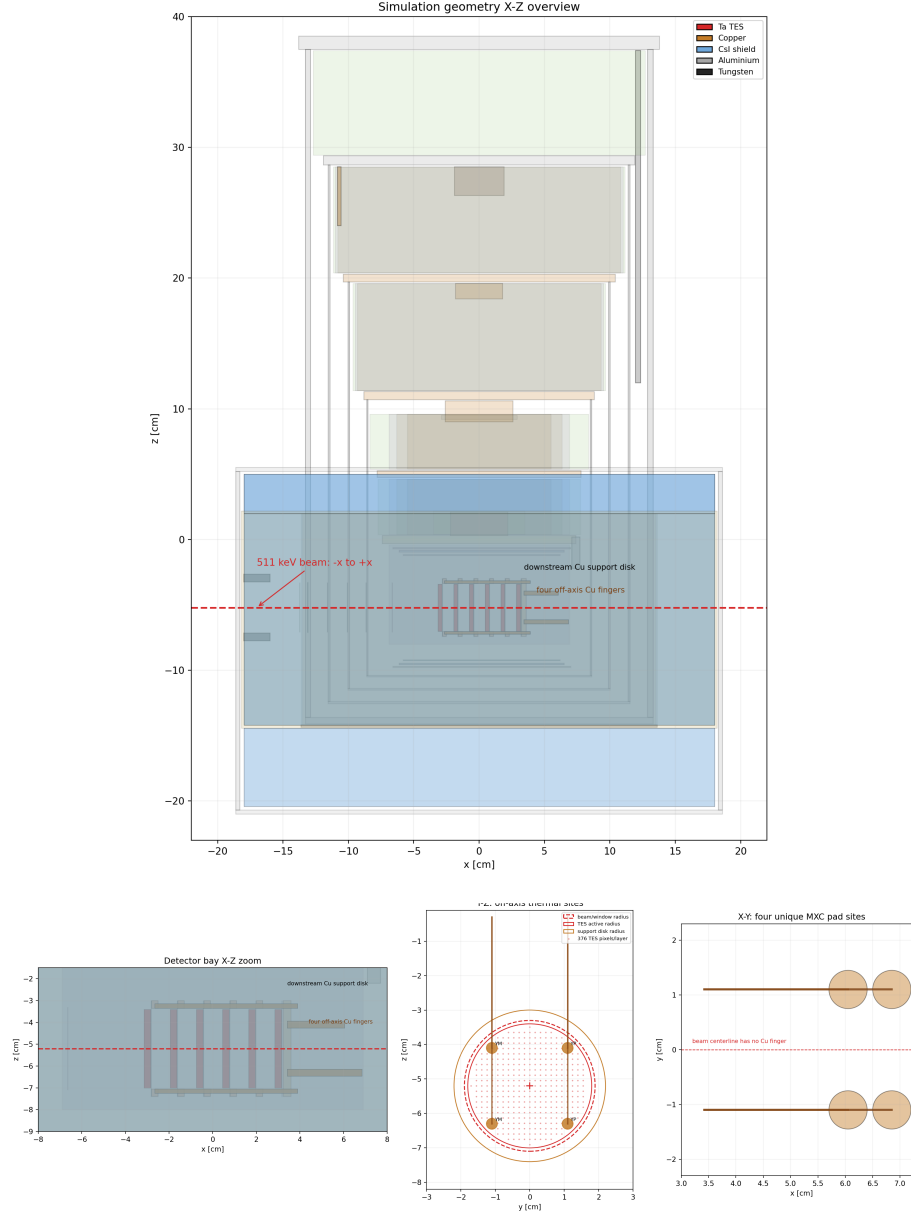


Figure 2: Two-dimensional views of the baseline detector/cryostat proxy mass model. Top: full X-Z overview of the DR tower, detector bay, active shield, and side-entry 511 keV beam path. Bottom: detector-bay detail showing the TES absorber stack, open central beam region, and off-axis Cu thermal links. These are simulation schematics; circular x-axis parts and side apertures are represented by MEGAlib-compatible proxy shapes rather than final CAD solids.

5. Laue optics and detector-coupled signal replay

The signal model follows the standard Laue-lens approach in which mosaic crystals diffract incident gamma rays in transmission and concentrate them onto a small focal detector [8, 9]. This is the same optical principle adopted in earlier focused gamma-ray telescope studies, where detector response must be evaluated together with the focused phase space rather than from lens effective area alone [10]. For the present 511 keV TES concept [7], the baseline optical configuration is a single-energy Ge(111) Laue ring with focal length 10 m, 25 square tiles of 18 mm side length, 30 arcsec mosaic FWHM, and active Ge mass 0.44063 kg. External XOP/CRYSTAL rocking-curve maps [11] set the Ge(111) diffraction probability and angular response. The effective-area normalization is the ring geometric area multiplied by the simulated fraction of diffracted photons that emerge through the detector entrance aperture, giving $A_{\text{eff}}(511 \text{ keV}) = 20.08476 \text{ cm}^2$.

The optical simulation is used as a source generator rather than as a standalone effective-area calculation. During transport, gamma rays that cross the nominal focal plane are recorded with their energy, position, direction, and production history. Only tracked diffracted photons that fall within the Be entrance-window aperture are passed downstream; analytic spot projections are retained only as diagnostics. This preserves the attenuation and scattering that occur while the diffracted photon exits the Ge crystal.

Figures 3 and 4 show the two levels of the optics-detector coupling: the radial optical constraint from the configured Laue ring to the focal plane, and the transverse distribution injected into the detector model.

The accepted focal-plane photons are transformed into the local side-entry detector coordinate system and replayed through the detector/cryostat mass model. This keeps the simulated entrance position and direction distribution, rather than replacing the optical calculation with an analytic focal spot. The focal-plane crossing table used here is therefore a transported optical-signal input: it records gamma rays from the 511 keV source simulation that cross the focal plane, not particles produced by atmospheric or cosmic-ray interactions in the upstream lens hardware.

An auxiliary upstream-hardware calculation was run to test that this boundary can be treated by the same Geant4/MEGAlib background machinery when needed. The calculation places an equal-volume and equal-mass Ge proxy for the active Laue crystals at the 10 m upstream lens position. The proxy corresponds to 25 square Ge tiles of 18 mm side length and 10.218801

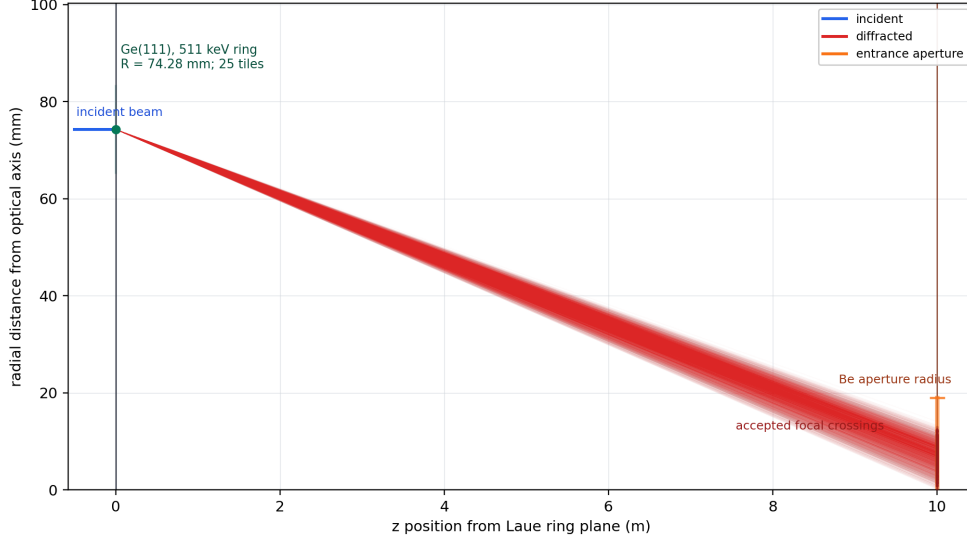


Figure 3: Data-constrained radial- z visualization of the baseline 511 keV Laue signal source. The Laue plane, Ge(111) ring radius and tile footprint, 10 m focal distance, and Be entrance-aperture radius are read from the current optical configuration. The diffracted fan terminates on tracked focal-plane crossings that satisfy the entrance-aperture gate. The plot collapses the azimuthal tile layout and is used only to visualize the transport-backed detector input.

mm thickness, giving active Ge volume 82.7723 cm^3 and mass 0.4406 kg . The atmospheric far-field source surface is not inherited from the local 60 cm detector source sphere. Instead, it is derived from the combined detector-plus-lens enclosing radius: the 10 m lens center plus the active-crystal volume bound gives 1008.8 cm, and the transport source surface is rounded upward to a radius of 1060 cm inside a 2500 cm world half-size. After low-statistics interface tests, production prompt and activation-production transports were run for the proxy with the same eight EXPACS/PARMA particle families (γ , n , $e^\pm$, p , α , and $\mu^\pm$) used in the detector-background chain. These runs define the full-particle transport and isotope-production boundary for the upstream mass proxy. The delayed component from the active-Ge proxy is isolated from the combined isotope-production records and replayed as a recorded-position decay source through the same detector-level selection; its selected-rate consequence is reported in Section 9.5.

For the focused point-source response, $F_0 = 10^{-4} \text{ cm}^{-2} \text{ s}^{-1}$ denotes the reference incident 511 keV photon flux. The quoted hard-window signal

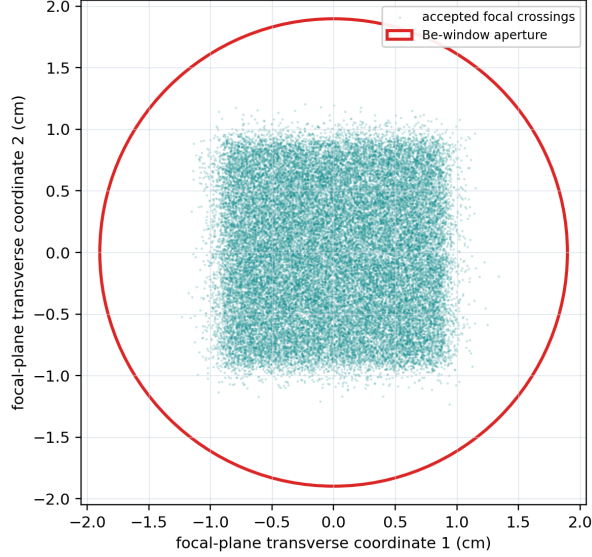


Figure 4: Tracked transverse focal-plane positions accepted by the Be entrance aperture and used for downstream detector replay. The red circle marks the Be-window aperture. This is the transported optical phase-space distribution at the detector input plane, not an analytic focal-spot projection.

response, $0.00118587 \text{ s}^{-1}$, is the selected W_{511} signal rate at that flux after detector transport, the reference atmospheric-transmission factor, and detector-level event selection. The source-flux response is the corresponding linear coefficient, $S_{W_{511}}/F_0 = 11.8587 \text{ s}^{-1}/(\text{ph cm}^{-2} \text{ s}^{-1})$. This reference normalization should be read as the baseline pointing case, not as a universal atmospheric transmission for all balloon pointings. Low-elevation or long-slant observations, including Galactic-center observations from northern mid-latitude balloon trajectories, should use a slant-column transmission model; such slant-column calculations remain boundary checks rather than replacements for the headline hard-window result.

6. Prompt atmospheric and delayed-activation source model

6.1. Prompt atmospheric source

The background source model is separated into prompt atmospheric transport and delayed activation so that source generation does not absorb assumptions that belong to detector selection. The prompt atmospheric

field is derived from EXPACS/PARMA. PARMA3.0 is an analytical model parameterized from PHITS atmospheric-shower calculations and benchmarked for terrestrial cosmic-ray fields, and EXPACS is its public tabulation/software interface [15]. PHITS supplies the underlying particle-transport calculation for cosmic-ray interactions in the atmosphere: primary cosmic rays interact with air nuclei, secondary hadrons, leptons, photons, and neutrons are transported through the atmospheric column, and the resulting particle fluences are parameterized by altitude, geomagnetic cutoff, solar modulation, energy, and direction. In this work EXPACS/PARMA provides the differential incident particle flux as a function of particle species, kinetic energy, zenith angle, geographic position, altitude, geomagnetic cutoff rigidity, and solar modulation. It is used only to construct the external radiation fields; detector response, vetoing, topology selection, and mission-time counting are applied later. The reference source definitions use latitude 34° , longitude 100° , altitude 38 km, vertical cutoff rigidity $R_c = 11.6$ GV, and the 2025-08-31 solar condition corresponding to $W = 118.3$.

The prompt source includes photons, neutrons, electrons, positrons, protons, alpha particles, negative muons, and positive muons. For each particle family, the EXPACS/PARMA differential spectrum is converted into a MEGALib far-field area source over the full zenith range $\theta = 0^\circ$ – 180° and full azimuth range $\phi = 0^\circ$ – 360° . The zenith dependence is represented by 20 differential equal- μ angular bins, where $\mu = \cos \theta$ and each bin subtends $\Delta\Omega = 0.6283185307$ sr. The first ten bins cover down-going particles ($0^\circ \leq \theta \leq 90^\circ$), and the last ten bins cover up-going or albedo-like particles ($90^\circ \leq \theta \leq 180^\circ$). Both hemispheres are retained in both the prompt transport and the activation-production transport, so upward albedo particles are not discarded at source construction.

Figure 5 shows the resulting full-sphere flux field.

The integrated source-plane fluxes are 4.79966, 0.462239, 0.199002, 0.116981, 0.112301, 0.0114871, 0.00496893, and 0.00557001 $\text{cm}^{-2} \text{s}^{-1}$ for γ , n , e^- , e^+ , p , α , μ^- , and μ^+ , respectively. The non-photon components therefore carry about 0.913 $\text{cm}^{-2} \text{s}^{-1}$, or 16% of the total full-sphere source-plane flux. To keep their detector and activation statistics from being limited by strict flux-proportional event counts, the non-photon particle families are generated with replicated Monte Carlo samples and then de-weighted in the rate normalization. These particles are transported through the detector/cryostat mass model with Geant4/MEGALib [18, 19]. The

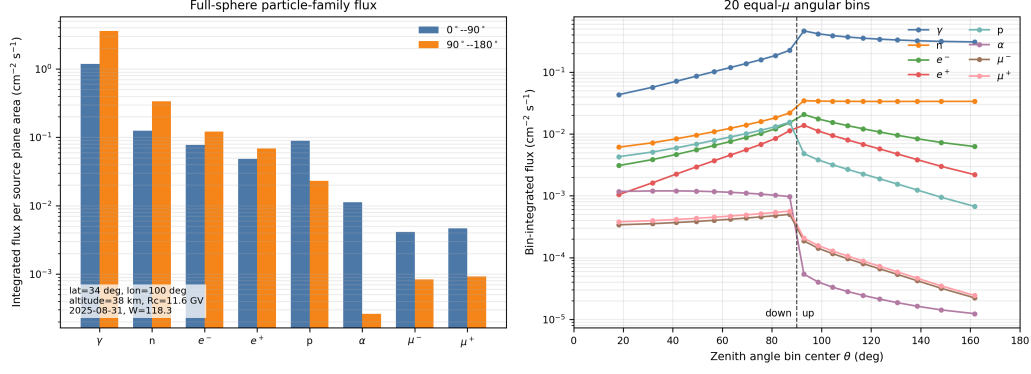


Figure 5: EXPACS/PARMA full-sphere atmospheric flux field used to generate the prompt and activation-production transport inputs. Left: energy- and solid-angle-integrated fluxes for the eight transported particle families, separated into down-going and up-going hemispheres. Right: bin-integrated flux in the 20 equal- μ differential zenith bins used by the MEGAlib far-field source. The plot is generated from the source-definition inputs for latitude 34°, longitude 100°, altitude 38 km, $R_c = 11.6$ GV, and $W = 118.3$.

prompt run and the activation-production run each generated 25,210,216 incident primary particles. Each particle species is normalized with its own transported exposure using

$$r_{\text{event},j} = \left(\sum_i \mathcal{T}_{ij} \right)^{-1}, \quad (1)$$

where $\mathcal{T}_{ij}$ is the transported exposure of subsample i for particle species j . This per-species normalization is required because photon and non-photon atmospheric streams have different sampling and exposure structures. The normalization check enforces $r_{\text{event},j} \sum_i \mathcal{T}_{ij} = 1$ for each particle species, so over-sampling changes the Monte Carlo variance but not the expected physical rate.

6.2. Delayed activation source

The activation-production stream is not counted as an immediate detector background. Instead, it records the isotope, material volume, and production position of radioactive products. These records are converted into a fixed day-15 radioactive population using production-decay buildup equations and ground-state half-life corrections checked against NUBASE2020 [16]. For

isotope k ,

$$A_k(t_{\text{flight}}) = R_k [1 - \exp(-\lambda_k t_{\text{flight}})], \quad \lambda_k = \frac{\ln 2}{t_{1/2,k}}, \quad (2)$$

where R_k is the production rate inferred from the activation-production transport and $t_{\text{flight}} = 15$ d for the reference day-15 state. The resulting fixed delayed activity used by the current fix5 chain is 85.449203 Bq.

The position information is not obtained from the aggregate isotope store alone. The standard isotope store records the volume name and isotope count, while the exact-position chain also uses a custom hook in the Geant4/MEGAlib stepping-action layer. At the same point where an isotope is stored during activation buildup, the hook writes a production-position record containing the logical volume, pre-step position (x, y, z) , isotope identifier $1000Z + A$, excitation energy, time, process, and track ancestry metadata. The fixed day-15 isotope population is then joined to these records by volume, isotope, and excitation state. This gives the delayed source access to the true simulated production coordinates rather than only to volume-integrated activities.

Figure 6 shows the resulting delayed-source position distribution after matching radioactive-production positions to the fixed day-15 activity population.

The activity-weighted distribution is dominated by CsI shield activation and by ^{128}I , with smaller contributions from Cu, Al, W, Mg, and Cs isotopes. The figure is a source-construction diagnostic, not a detector-count map; detector response is applied only after the delayed decays are transported through the common geometry and event selection.

The need for exact-position sampling is also testable from the same production table. Delayed decays are emitted locally, so the downstream attenuation, active-shield veto probability, and event topology depend on where the isotope was produced. Charged hadrons and muons have different energy-loss, stopping, and capture histories from neutrons, and therefore need not produce an exchangeable material or coordinate distribution. To quantify this, the weighted production rows were grouped by incident particle family and by material category. For two families a and b , we compare the activity-normalized category distributions with

$$D_{\text{TV}}(a, b) = \frac{1}{2} \sum_c |f_{a,c} - f_{b,c}|, \quad (3)$$

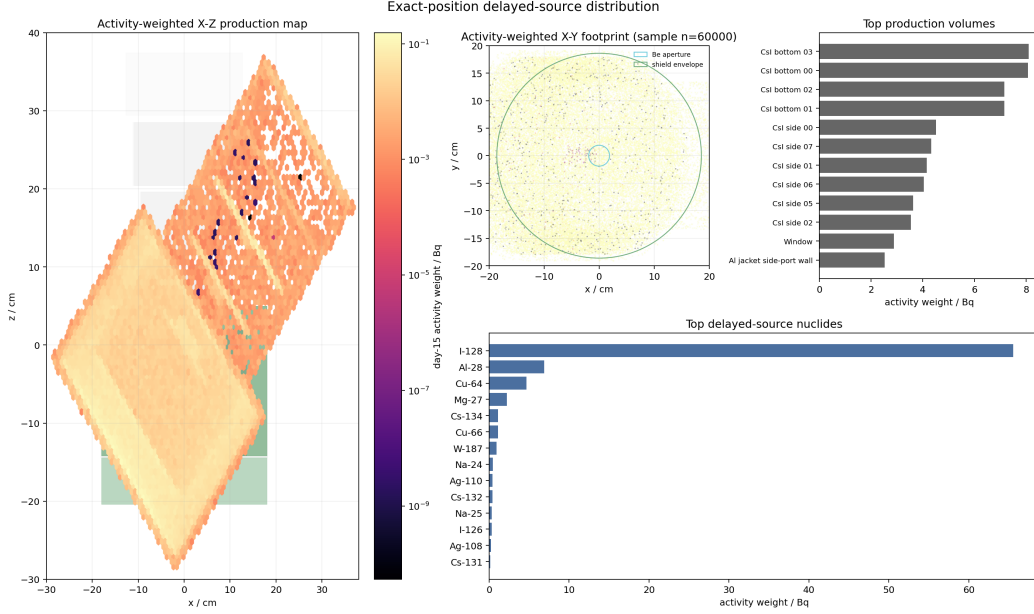


Figure 6: Exact-position delayed-source sample constructed from true radioactive-isotope production positions in the current fix5 chain. Left: activity-weighted X-Z map of the recorded production positions overlaid on the detector/cryostat proxy geometry. Top middle: activity-weighted X-Y footprint sampled from the same distribution, with the Be aperture and shield envelope shown for scale. Right and bottom: leading production volumes and nuclides after fixed day-15 activity weighting. The activity weights in the matched production-position table sum to 85.449203 Bq.

where $f_{a,c}$ is the fraction of delayed activity from incident family a produced in material category c .

Figure 7 gives the result of this test.

In the fixed day-15 table, the activities are 82.522 Bq for neutron-induced products, 2.752 Bq for μ^- -induced products, 0.147 Bq for proton-induced products, and 0.0223 Bq for α -induced products, with smaller μ^+ and e^+ contributions. The neutron-induced activity remains dominated by the CsI active shield, while the μ^- component is qualitatively different: Cu/support material, Al shells, and W/collimator material account for 32.2%, 28.5%, and 20.5% of the μ^- -induced activity, respectively. The fixed source contains 0.986 Bq in W/collimator volumes, or 1.154% of the fixed delayed activity. Because the selected-event audit finds zero W/collimator-origin selected delayed events in the hard-window sample, this source-level W activation does not become a dominant selected-rate component. A volume-integrated or

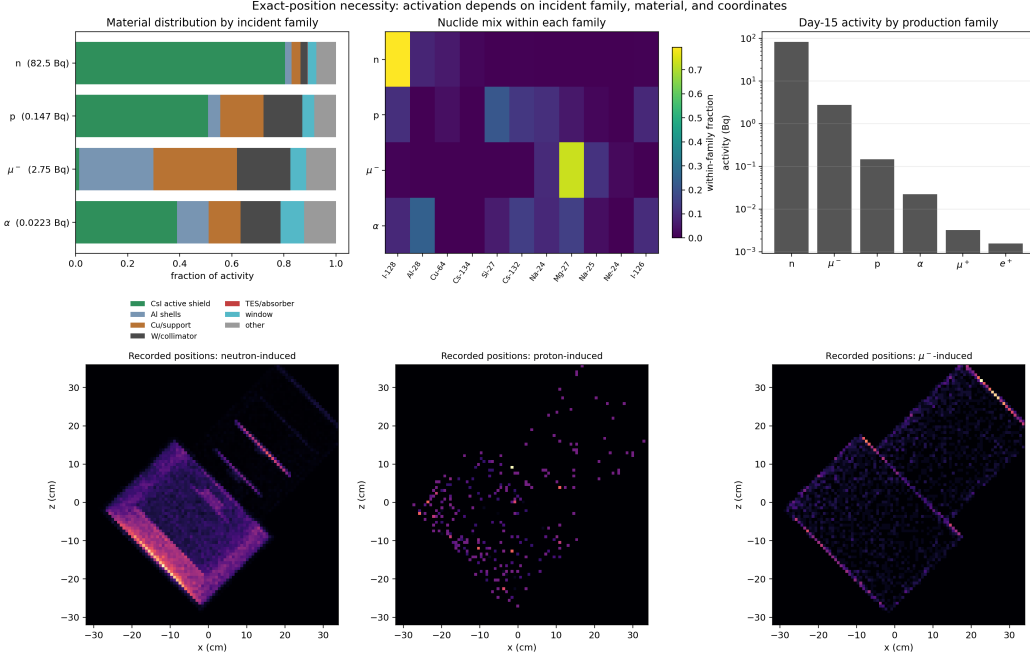


Figure 7: Data-backed necessity test for exact-position delayed-source sampling. The upper panels compare the day-15 weighted production table by incident particle family: material fractions, nuclide mix, and total activity. The lower panels show the recorded X-Z production-position maps for neutron-, proton-, and μ^- -induced products. The distributions are not exchangeable: neutron-induced activity is dominated by CsI and ^{128}I , while μ^- -induced activity is concentrated in Al/Cu/W service and shield material and is dominated by Mg/Na/Ne products.

axisymmetric radial source could preserve total activity, but it would remove incident-family, material, nuclide, and coordinate correlations. The exact-position construction keeps them in the source distribution before detector transport.

The delayed-decay source is launched from sampled production positions rather than from an axisymmetric radial profile. Let rows j in the weighted production table have activity weights w_j with total activity $A = \sum_j w_j$. The exact-position approximation draws M production positions with replacement using probabilities $p_j = w_j/A$, and assigns each sampled point source flux A/M . The expected activity assigned to row j is therefore

$$\mathbb{E}\left[N_j \frac{A}{M}\right] = M p_j \frac{A}{M} = w_j, \quad (4)$$

so the estimator is activity preserving and unbiased for the weighted spatial distribution. Here M denotes the number of equal-activity sampled point sources used to approximate the distribution; it is a sampling size, not the number of physical isotope-production rows. The source used for the primary fix5 rate result uses $M = 50000$ sampled point sources drawn from the full weighted production table with seed 260613. These sampled points preserve the total fixed activity exactly at the source-definition level and are replayed through the same detector/cryostat geometry used for the prompt stream. The nominal fix5 delayed calculation transported 10^6 delayed decays, with Cosima live time 11649.564832 s. This live time is a transport record for the generated decay sequence and is not used as the physical balloon exposure.

Table 4 collects the source layers used by the current fix5 detector-coupled calculation. It is placed in the source-model section because it defines the incident and delayed radioactive populations before detector vetoes and topology selections are applied.

This exact-position representation is not an exhaustive source with one sampled point source for every production row. It is a statistically sampled representation whose adequacy is evaluated downstream in Section 9.3. The primary fix5 rate result uses the transported $M = 50000$, seed-260613 sampling case, with source-inventory and selected W/collimator audits used to check the delayed normalization.

7. Detector event selection and veto definition

The analysis window is

$$W_{511} = [510.58, 511.42] \text{ keV}, \quad (5)$$

corresponding to $511 \text{ keV} \pm 420 \text{ eV}$. This window is matched to a sub-keV TES response assumption and is narrow enough to suppress continuum background while retaining most of the line response. The quoted flux thresholds therefore apply to a line whose intrinsic width remains small compared with this analysis window after detector response; broader source lines would require convolving the source line profile with the detector response and re-optimizing or widening the analysis window.

The detector selection has two explicitly separated veto layers. First, an active anticoincidence gate is applied on a common detector time axis. For a candidate group c , formed from event instances separated by no more than

$\tau = 1 \mu\text{s}$, the TES energy and active-shield energy are

$$E_{\text{TES}}^{(c)} = \sum_{j \in c} E_{\text{TES},j}, \quad E_{\text{sh}}^{(c)} = \sum_{j \in c} E_{\text{sh},j}. \quad (6)$$

A candidate enters the hard line window if $E_{\text{TES}}^{(c)} \in W_{511}$, and it passes the active shield if

$$E_{\text{sh}}^{(c)} < E_{\text{veto}}, \quad E_{\text{veto}} = 50 \text{ keV} \quad (7)$$

for the baseline CsI configuration. The shield sum is evaluated from the CsI active-shield volumes in post-processing. Native detector trigger/veto blocks are present in the geometry description, but the quantitative rate definition is this common post-processing definition, so prompt, delayed, and focused-signal streams use the same coincidence window and threshold.

The second layer is a Compton/field-of-view consistency test on the TES hits that survive the active-shield gate. Single-pixel TES events are kept. For a two-hit event, both possible first-scatter orders are tested. For an assumed order with first deposited energy E_1 and remaining scattered-photon energy E_{sc} , the kinematic angle is computed from

$$\cos \theta_C = 1 - m_e c^2 \left(\frac{1}{E_{\text{sc}}} - \frac{1}{E_1 + E_{\text{sc}}} \right). \quad (8)$$

The event is kept if either order gives a physically valid first-scatter cone that intersects the side Be-window disk. For higher multiplicities the current implementation enumerates candidate hit orders up to six TES hits, selects the best sequence using the usual Compton sequence residual

$$Q = \sum_i (\cos \theta_{C,i} - \hat{\mathbf{u}}_{i-1,i} \cdot \hat{\mathbf{u}}_{i,i+1})^2, \quad (9)$$

and then applies the same first-cone intersection test. If no valid sequence is found, the event is classified as a reconstruction reject but is retained under the present retained-reconstruction-reject policy. The hard-window result therefore does not gain sensitivity from rejecting ambiguous reconstruction failures. This policy is not a conservative lower bound on the net sensitivity. In the current W_{511} sample, a diagnostic alternative that discards the retained reconstruction-reject class would retain 59.2% of the focused signal and 66.8% of the selected background, reducing the simple $S/\sqrt{B}$ figure to 0.724 of the baseline value. The baseline selection therefore treats ambiguous multi-hit

events as a reconstruction-model boundary that must be validated, not as a sensitivity gain from aggressive event rejection.

The detector-level time-axis construction used to validate this veto book-keeping is a Poisson superposition of independent streams. For stream k with event-catalogue rates r_{kj} and total rate $R_k = \sum_j r_{kj}$, a validation draw uses

$$N_k \sim \text{Pois}(R_k T), \quad (10)$$

selects event records with probabilities r_{kj}/R_k , assigns uniform times in $[0, T]$, sorts all streams together, and forms τ -coincidence candidate groups. This draw tests the grouping and accidental-coincidence bookkeeping; the mission-time calculation below uses the corresponding expectation values instead of adding a new random draw to every mission bin.

Figure 8 summarizes the selected rates after these veto layers and previews the mission-time fold defined in Section 8, while Figure 9 gives the aperture geometry used by the Compton/FoV consistency test.

8. Mission-time trajectory fold and significance calculation

After detector selection is fixed, the selected day-15 rates are folded through a reference mission-time trajectory. This layer is a rate-level correction: no new Geant4/MEGAlib transport is run in each mission-time bin. The time grid has 81 points from 0 to 20 d with 0.25 d spacing; endpoint points carry half-widths in the count integration. The trajectory is a synthetic reference profile, not balloon telemetry. For time-grid point t_i in days,

$$h_i = h_0 + \Delta h \sin \left[\frac{2\pi(t_i - 15)}{5} \right], \quad (11)$$

$$\varphi_i = \varphi_0 + \Delta\varphi \sin \left[\frac{2\pi(t_i - 15)}{7} \right], \quad (12)$$

$$\ell_i = \ell_0 + \Delta\ell \sin \left[\frac{2\pi(t_i - 15)}{9} \right], \quad (13)$$

where $h_0 = 38.75$ km, $\Delta h = 2.5$ km, $\varphi_0 = 34^\circ$, $\Delta\varphi = 0.25^\circ$, $\ell_0 = 100^\circ$, and $\Delta\ell = 0.25^\circ$. The resulting ranges are 36.25–41.25 km in altitude, 33.75° – 34.25° in latitude, and 99.75° – 100.25° in longitude.

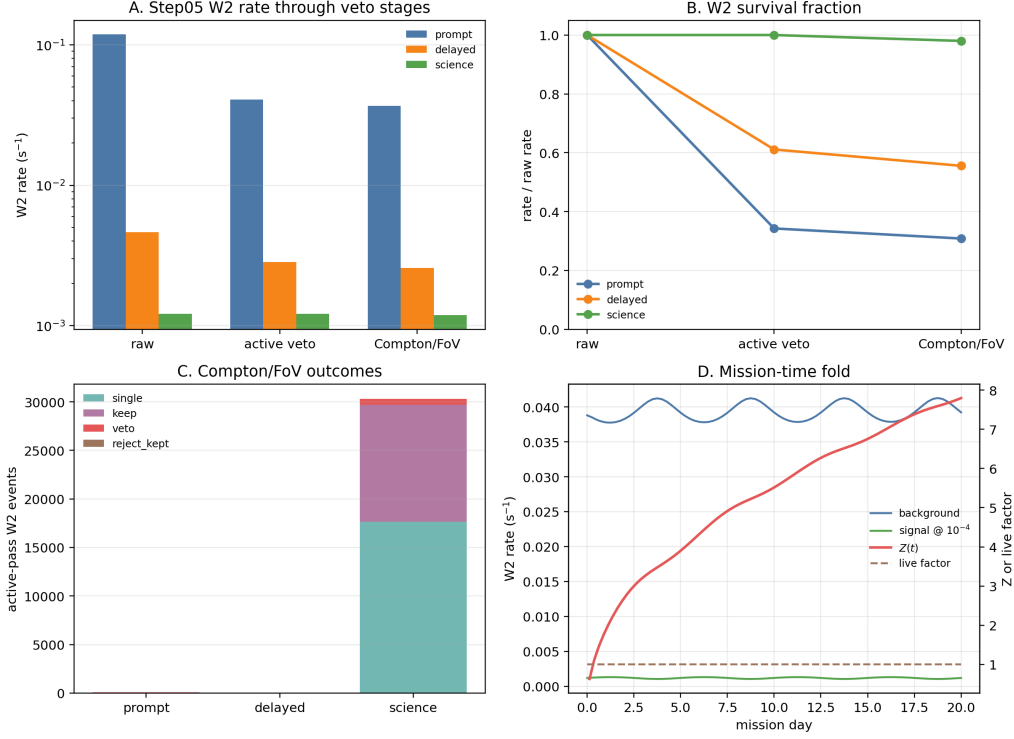


Figure 8: Detector-selection and mission-time fold diagnostics for the current fix5 $M = 50000$ exact-position delayed-source chain. Panel A shows the W_{511} pre-veto, active-veto-pass, and Compton/FoV-pass rates for prompt background, delayed activation, and the focused signal scaled to $F_0 = 10^{-4} \text{ cm}^{-2} \text{ s}^{-1}$. Panel B shows the corresponding survival fractions. Panel C shows the Compton/FoV classifications after the active-shield gate. Panel D shows the mission-time rate fold for the reference point-source case, including the analytic accidental live factor.

The atmospheric depth and the 511 keV transmission scale are

$$X(h_i) = X_{38} \exp\left[-\frac{h_i - 38 \text{ km}}{H}\right], \quad X_{38} = 3.86510 \text{ g cm}^{-2}, \quad H = 6.8 \text{ km}, \quad (14)$$

$$T_i = \exp[-\mu_{\text{eff}} X(h_i)], \quad \mu_{\text{eff}} = -\frac{\ln T_{15}}{X(h_0)}, \quad (15)$$

with $T_{15} = 0.7390423888027$. Thus the focused-science scale is $f_{\text{atm},i} =$

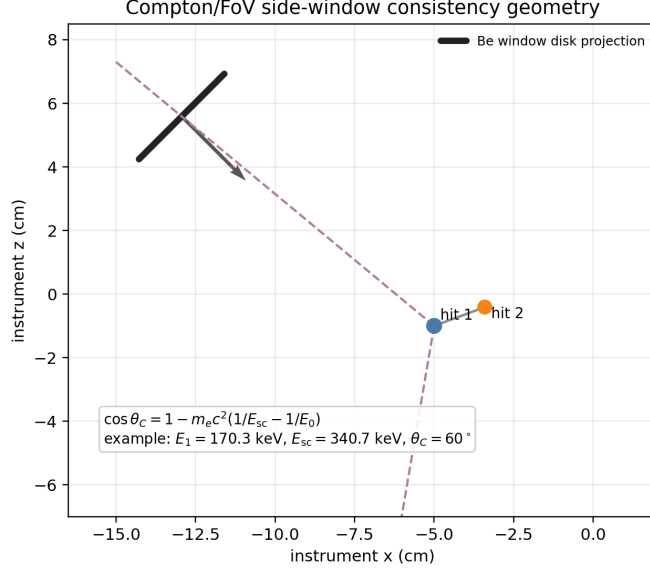


Figure 9: Geometry used by the Compton/FoV consistency test. The Be-window disk projection is generated from the current side-entry disk center, normal, and radius used by the detector-coordinate definition. Two-hit and higher-multiplicity candidates are tested by asking whether the first-scatter Compton cone intersects this aperture, not by requiring consistency with a fitted focused spot.

T_i/T_{15} . The prompt atmospheric source scale is

$$R_{c,i} = 11.0 - 0.08(\varphi_i - \varphi_0) + 0.03(\ell_i - \ell_0), \quad (16)$$

$$f_{p,i} = \exp \left[\frac{X(h_i) - X(h_0)}{30 \text{ g cm}^{-2}} \right] \left(\frac{11.0}{R_{c,i}} \right)^{0.2}. \quad (17)$$

The same scalar production driver is used for delayed activation before isotope-by-isotope decay buildup is solved. For a delayed-production element a with decay constant λ_a and fixed day-15 activity $A_{a,15}$,

$$P_{a,15} = \frac{A_{a,15}}{1 - \exp(-\lambda_a t_{15})}, \quad (18)$$

$$\frac{dN_a}{dt} = P_{a,15} f_p(t) - \lambda_a N_a, \quad A_a(t) = \lambda_a N_a(t). \quad (19)$$

The numerical fold anchors the solution back to $A_{a,15}$ at day 15 and sums the activity over the nuclide/material/position production elements.

This delayed-activity treatment assumes that the normalized activation-product distribution is unchanged over the small trajectory range, and that the trajectory changes only the scalar production driver. The verification described here proves closure of that model but does not prove physical invariance of the production distribution. If dedicated activation transports at the trajectory center and extrema show material or nuclide redistribution, the scalar driver should be replaced by interpolation over production vectors,

$$P_a(t_i) = \sum_g w_g(h_i, \varphi_i, \ell_i, R_{c,i}) P_a^{(g)}, \quad \sum_g w_g = 1, \quad (20)$$

followed by the same decay ODE.

A lightweight verification recomputed the trajectory formula at the center point and at the two altitude extrema. At day 15, $h = 38.75$ km, $T = 0.739042388803$, $f_p = 1$, and the delayed-activity scale is 1. At the high-altitude probe, day 1.25 gives $h = 41.25$ km, $T = 0.811094979198$, $f_p = 0.965181809612$, and delayed-activity scale 0.913070839359. At the low-altitude probe, day 3.75 gives $h = 36.25$ km, $T = 0.646122572391$, $f_p = 1.05298763053$, and delayed-activity scale 1.02178977033. The maximum formula delta at these probes is zero in the verification. Reconstructing the W_{511} prompt, delayed, background, and signal rates from the day-15 rates and these scales also gives zero relative delta, and the final cumulative source counts, background counts, and Z_{20d} reproduce the summary values exactly.

A second smoke test probes the physical part of this approximation at the particle-family remixing level. Local PARMA spectra were regenerated at the low-altitude, day-15, and high-altitude trajectory probes and were compared with the full-statistics activation-buildup isotope store grouped by incident particle family. Reweighting the day-15 isotope/volume activity contributions by the PARMA particle-family scale changed the normalized nuclide-volume activity distribution by at most 5.72×10^{-3} in total-variation distance. The largest PARMA particle-family distance was 6.76×10^{-4} , the largest particle-energy-bin distance was 9.05×10^{-3} , and the matched ground-state activity fraction in the parsed production table was 0.9929. This supports using a scalar delayed production/activity fold for the present small trajectory range at the particle-family reweighting layer. It is not a full proof of distribution invariance: same-particle energy/angle-dependent activation cross sections and position-transport shifts would require dedicated low-, reference-, and high-altitude activation-production transports.

In the mission-time fold, the selected day-15 rates are folded over the trajectory grid as

$$B_i = L_i (B_{p,15} f_{p,i} + B_{d,15} f_{d,i}), \quad (21)$$

$$S_i = L_i S_{15} f_{\text{atm},i}, \quad (22)$$

$$L_i = \exp[-(R_{p,i} + R_{d,i})\tau], \quad (23)$$

where $f_{p,i}$ is the prompt atmospheric scale, $f_{d,i}$ is the delayed-activity scale, $f_{\text{atm},i}$ is the 511 keV atmospheric transmission scale, and L_i is the accidental coincidence live factor. Here $R_{p,i}$ and $R_{d,i}$ are the prompt and delayed event rates entering the coincidence grouping at that mission point.

The time-dependent significance is computed as a cumulative counting statistic. For a reference flux F_0 , the cumulative significance is

$$Z(t_m) = \frac{N_s(t_m)}{\sqrt{N_b(t_m)}}, \quad (24)$$

where

$$N_s(t_m) = \sum_{i \leq m} S_i \Delta t_i, \quad N_b(t_m) = \sum_{i \leq m} B_i \Delta t_i \quad (25)$$

are the time-integrated selected signal and background counts after mission-time scaling and live-factor corrections. For comparison among analysis variants, the 3σ flux threshold is scaled as

$$F_{3\sigma}(t_m) = F_0 \frac{3}{Z(t_m)}. \quad (26)$$

This Gaussian counting convention is used consistently for the hard-window results and secondary comparisons. For the primary 20-d hard-window result, the folded counts are about 1.09×10^5 background counts and 2.0×10^3 signal counts at F_0 , so this approximation is used only in the high-count selected-rate regime. The isolated upstream Ge-proxy result with zero selected events is reported separately with a Poisson zero-count upper limit. The detector-selection summary also records a day-15 direct expectation. The primary result does not use that constant-rate direct value; it uses the mission-time fold and the final cumulative significance. The distinction is small for the present hard-window case, but keeping it explicit is necessary for comparing analysis configurations.

9. Results

9.1. Primary hard-window sensitivity

Table 5 gives the primary hard-window result. In the current fix5 exact-position chain and at $F_0 = 10^{-4} \text{ cm}^{-2} \text{ s}^{-1}$, the selected day-15 rates are

$$B_{W_{511}} = 0.0392162 \text{ s}^{-1}, \quad S_{W_{511}} = 0.00118587 \text{ s}^{-1}. \quad (27)$$

The mission-time fold gives mean selected rates

$$\bar{B}_{W_{511}} = 0.0393546 \text{ s}^{-1}, \quad \bar{S}_{W_{511}} = 0.00117748 \text{ s}^{-1}. \quad (28)$$

The resulting 20-d significance is $Z_{20d} = 7.79950$, with a 3σ detection time of 2.50573 d. The 20-d 3σ flux threshold is $3.84640 \times 10^{-5} \text{ cm}^{-2} \text{ s}^{-1}$.

The hard-window row uses the narrow energy window, active-shield veto, topology/FoV consistency selection, and mission-time fold. The previous-authority row shows the replacement comparison used by the fix5 promotion decision: the total selected background decreases by 37.7%, the signal response is preserved at the 0.4% level, and the 20-d flux threshold decreases by 21.4%. Spatial gates, 45° line-of-sight folds, and fixed-template annular likelihoods remain archived boundary tests and post-processing analyses, not finalized nuisance-profile likelihood results.

9.2. Background composition

The selected fix5 exact-position W_{511} background is dominated by prompt atmospheric events. The selected stream split is

$$B_{\text{prompt}} = 0.0366410 \text{ s}^{-1}, \quad B_{\text{delayed}} = 0.00257520 \text{ s}^{-1}, \quad (29)$$

corresponding to 93.43% and 6.57% of the total selected hard-window background. The selected W/collimator-origin delayed audit matches the delayed selected events back to the sampled exact-position source inventory and finds zero W/collimator-origin selected events. Table 6 lists the selected-rate components and the W/collimator activation checks.

9.3. Fix5 source-inventory and delayed-activation audits

Table 7 summarizes the source-inventory and selected-event checks attached to the current fix5 exact-position closure. These checks replace the older v3p5 M-sampling sweep as the promotion evidence for the present

geometry. The previous sweep remains a useful method check, but the current pass/fail decision is based on the full-statistics $M = 50000$, seed-260613 fix5 source, its normalization audit, and the downstream selected-event W/collimator audit.

The fixed source contains 251,681 weighted production-position rows and 50,000 equal-activity point-source blocks. The source text sums to 85.4492025 Bq, with a relative source-text flux discrepancy of 4.54×10^{-10} . The NUBASE ground-state correction lowers the raw delayed activity from 87.4833 to 85.4492 Bq. W/collimator volumes carry 0.986080 Bq of source-level activity, but the selected-event audit finds zero W/collimator-origin selected delayed events among 30 selected delayed events.

9.4. Fix5 full-statistics promotion against the previous authority

Table 8 summarizes the full-statistics promotion comparison. The current fix5 hard-window result is compared with the previous v3p5 exact-position authority using the same selected-rate definition. The old new-geometry prompt and delayed totals are not used as a pass/fail reference because the benchmark alignment audit marks that chain as not directly aligned with the current selection and normalization.

9.5. Upstream optics mass-proxy boundary

The upstream active-Ge proxy produces two isolated day-15 radioactive components, ^{70}Ga and ^{73}Ga , with total activity 0.425674 Bq. A 20,000-decay delayed replay through the detector geometry and detector-level selection produced no selected W_{511} events. The corresponding direct selected rate is zero, with a 95% zero-count upper rate of $1.25 \times 10^{-4} \text{ s}^{-1}$. This is about 0.32% of the current fix5 exact-position W_{511} background rate, 0.0392162 s^{-1} , so the isolated delayed Ge-proxy component does not change the quoted hard-window sensitivity at the current precision. The prompt self-background from upstream hardware is still not folded into the primary budget because the combined prompt simulation also contains the ordinary detector/cryostat atmospheric prompt background and needs source separation or subtraction. Explicit lens-tile supports remain a separate design-level systematic.

10. Discussion

The main result is that the current fix5 hard-window exact-position chain gives a 20-d 3σ point-source threshold of $3.85 \times 10^{-5} \text{ cm}^{-2} \text{ s}^{-1}$ for the modeled

focused TES line spectrometer. This threshold is obtained without using the focused-spot spatial analysis as the primary result. It follows from three combined effects: optical concentration, a sub-keV energy window, and a detector-coupled veto/topology selection that suppresses prompt and delayed backgrounds in the same selection space as the signal.

This calculation should be compared with existing 511 keV instruments as a pointed, narrow-line, compact-source estimate, not as an all-sky imaging forecast. INTEGRAL/SPI and COSI provide the reference measurements for mapping the diffuse 511 keV sky [2–5]. NASA and HEASARC mission pages describe COSI as a wide-field 0.2–5 MeV Ge Compton telescope; HEASARC lists the Ge detector energy resolution as 6.0 keV at 511 keV [23, 24]. The TES Laue-lens concept considered here is complementary: it trades sky coverage for a concentrated beam and a sub-keV line window. The relevant comparison class is compact 511 keV targets, central core searches, and transient follow-up observations with known or constrained source positions. The same focused aperture is not a substitute for measuring the full diffuse bulge and disk luminosity, because it samples only a small solid angle.

The exact-position delayed-source treatment addresses a specific approximation in radial-profile compression. Although delayed activation contributes only about 6.57% of the selected hard-window background in the present chain, it is spatially and materially structured. Preserving sampled production positions prevents artificial azimuthal or radial redistribution from becoming an unquantified systematic. The current result uses the transported $M = 50000$, seed-260613 fix5 sampling case, with the fixed-source inventory, source-card flux closure, and selected W/collimator audit used as the delayed-normalization checks.

The background composition points to the first engineering sensitivities to test, rather than to a completed optimization. The selected hard-window rate remains dominated by prompt atmospheric streams, so shield geometry, side-entry collimation, material near the focal plane, topology selection, and multi-pixel reconstruction are the most direct handles on the present background. Delayed activation is smaller but concentrated in identifiable nuclides and volumes; the current W/collimator audit is important because the redesigned passive material adds tungsten, but the selected-event rate shows no W/collimator-dominated delayed component.

The selected-rate result has four immediate boundaries. First, the detector/cryostat model is an engineering proxy, not a final flight CAD model. Second, quantitative active-shield rejection is applied in the common detector-

level post-processing selection rather than by production-time trigger/veto filtering. Third, the mission-time calculation is a rate-level fold, not a new particle transport at each trajectory time bin. Fourth, the custom Compton/FoV veto is not yet a full Revan/Mimrec reconstruction, and the retained reconstruction-reject class is a baseline acceptance choice rather than a conservative sensitivity bound.

More generally, the tabulated 3σ flux thresholds are statistical selected-rate thresholds for the current source, mass, and response model. They do not include a systematic-uncertainty envelope for atmospheric-flux normalization, activation cross sections or physics-list choices, material and geometry tolerances, detector energy-response uncertainty, veto-threshold calibration, or reconstruction-efficiency systematics.

The upstream Laue-lens calculation in Section 9.5 should be read as a boundary calculation for an equal-volume/equal-mass active-Ge proxy, not as a finalized optics-background budget. It completes full-particle prompt and activation-production transport for the proxy, and the isolated Ge-proxy delayed component gives zero selected W_{511} events in the present transport. The prompt self-background contribution is still not part of the primary detector-background budget because it needs source separation or subtraction from the ordinary detector/cryostat prompt background, and explicit lens support hardware is not yet modeled.

Spatial analyses such as r_{90} and multi-annulus likelihood estimates are diagnostic cross-checks, but they are not yet a finalized nuisance-profile likelihood. The promotion comparison in Section 9.4 is instead a same-selection comparison between the current fix5 closure and the previous v3p5 exact-position authority. The historical old new-geometry prompt and delayed totals are reported only as context because their benchmark-alignment audit is not directly comparable to the current selection and source-surface normalization. Thus the defensible replacement claim is against the v3p5 authority: fix5 lowers total selected background, preserves the signal response, improves Z_{20d} , lowers $F_{3\sigma}(20d)$, and does not introduce a W/collimator delayed red flag.

Before this rate study can support a flight-performance statement, four validation blocks remain: a Revan/Mimrec cross-check of the retained multi-hit subset; full-statistics optics self-background transport with explicit lens tiles and support hardware; a final cryostat/DR geometry and detector-response update, including TES saturation and pile-up constraints; and a spatial-spectral likelihood that profiles background nuisance parameters.

11. Conclusions

We have presented a detector-coupled background and sensitivity study for a balloon-borne pointed 511 keV TES Laue-lens telescope. The current fix5 chain couples a Ge(111) Laue optical response to a TES focal-plane mass model, prompt atmospheric transport, delayed activation, active-shield vetoing, Compton/FoV topology selection, mission-time rate folding, and counting-significance evaluation. In the primary 510.58–511.42 keV hard window, the day-15 selected rates are $B \simeq 3.92 \times 10^{-2} \text{ s}^{-1}$ and $S \simeq 1.19 \times 10^{-3} \text{ s}^{-1}$ at $10^{-4} \text{ cm}^{-2} \text{ s}^{-1}$; the mission-time fold gives $Z_{20\text{d}} \simeq 7.80$ and $F_{3\sigma}(20\text{d}) \simeq 3.85 \times 10^{-5} \text{ cm}^{-2} \text{ s}^{-1}$. Relative to the previous v3p5 authority, this lowers the selected hard-window background to 0.623 and the 20-d flux threshold to 0.786 while preserving the signal response. The source-inventory and selected W/collimator audits show that the delayed normalization is closed and that W activation is not a dominant selected-rate component. The result provides a detector-coupled selected-rate basis for continued evaluation of focused TES spectroscopy as a complementary 511 keV technique for compact-source and central-core searches, while identifying prompt atmospheric background suppression and reconstruction validation as the main next validation tasks.

Data and software availability

Before publication, a versioned public repository or institutional archive will provide the geometry and source definitions, random seeds, analysis and figure-generation scripts, LaTeX source, and derived validation products needed to reproduce the tables and figures in this paper. The archive will include machine-readable summaries and human-readable validation reports for the rate calculations, convergence checks, figure source tracing, and bibliography verification. Full raw transport outputs, compressed Cosima event files, and large intermediate event lists are substantially larger than the manuscript archive; they will either be deposited as a separate large-data package or represented in the archive by generation commands, configuration files, checksums or summary metadata, and derived validation products.

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Table 4: Background-source layers used by the current fix5 detector-coupled chain. The table defines the source construction only; vetoes, topology selections, and the hard line window are applied later in Section 7.

Layer		Physical role	Current implementation
Prompt atmospheric transport	at-	Direct detector hits from atmospheric secondaries	EXPACS/PARMA full-sphere source definitions at 34° latitude, 100° longitude, 38 km altitude, $R_c = 11.6$ GV, and $W = 118.3$; γ , n , $e^\pm$, p , α , and $\mu^\pm$ over 20 equal- μ differential zenith bins including both down-going and up-going hemispheres; 25,210,216 generated primaries.
Activation production	pro-	Radioisotope production and production-position history	Same atmospheric fields transported as a buildup stream, again retaining both hemispheres and all eight particle families; 25,210,216 generated primaries.
Fixed population	delayed	Radioactive activity after continuous exposure to the adopted day-15 state	Production-decay buildup with NUBASE-checked ground-state half-lives; total fixed activity 85.449203 Bq.
Exact-position decay transport		Delayed decay particles emitted from sampled production locations	$M = 50000$ equal-activity sampled point sources from the full weighted production table, seed 260613; 10^6 transported delayed decays.

Table 5: Primary fix5 sensitivity and previous-authority comparison. The fix5 exact-position hard-window row is the primary rate result. The previous v3p5 row is retained only to show the promotion comparison; it is not a second current geometry.

Case	Base	$B_{W_{511}}$ (cps)	$S_{W_{511}}$ at F_0 (cps)	Z_{20d}	$T_{3\sigma}$ (d)	$F_{3\sigma}(20d)$
Fix5 hard window	fix5 exact-position	0.0392162	0.00118587	7.79950	2.50573	3.84640×10^{-5}
Previous authority	v3p5 exact-position	0.0629804	0.00118117	6.13039	4.77766	4.89365×10^{-5}
Step05 constant-rate check	fix5 day-15 direct	0.0392162	0.00118587	7.87187	2.90480	3.81104×10^{-5}

The Step05 direct row is a constant-rate diagnostic before the mission-time fold; the promoted result is the fix5 cumulative time-dependent row. Archived spatial and slant-column rows are diagnostics and are not used to improve the headline hard-window threshold.

Table 6: Selected W_{511} background composition in the current fix5 exact-position chain. Percentages are relative to the total W_{511} background unless otherwise stated.

Component	Rate (cps)	Fraction
All prompt atmospheric streams	0.0366410	93.43%
All delayed activation streams	0.00257520	6.57%
W/collimator-origin selected delayed events	0	0% of delayed selected events
W/collimator fixed-source activity	0.986080 Bq	1.154% of fixed delayed activity

Table 7: Current fix5 delayed-source inventory and selected-event audits.

Check	Value	Status	Scope	Interpretation
Exact-position source blocks	$M = 50000$, seed 260613	PASS	source construction	Full-statistics fix5 delayed source
Weighted source inventory	251,681 rows, 85.449203 Bq	PASS	source normalization	Activity-preserving sampled source
Source text flux closure	relative delta 4.54×10^{-10}	PASS	source serialization	No source-card normalization drift
Ground-state correction	87.4833 $\rightarrow$ 85.4492 Bq	PASS	delayed activity	NUBASE ground-state half-life correction applied
Selected W/collimator audit	0 W/collimator selected events	PASS	detector-selected delayed events	W activation is not a dominant selected-rate component

Table 8: Fix5 full-statistics promotion comparison against the previous v3p5 exact-position authority. Ratios below unity are improvements for background and flux threshold; ratios above unity are improvements for signal retention and significance.

Metric	fix5	previous v3p5	fix5/v3p5
Day-15 total $B_{W_{511}}$ (cps)	0.0392162	0.0629804	0.6227
Delayed $B_{W_{511}}$ (cps)	0.00257520	0.00389764	0.6607
Signal $S_{W_{511}}$ at F_0 (cps)	0.00118587	0.00118117	1.0040
Cumulative Z_{20d}	7.79950	6.13039	1.2723
Cumulative $F_{3\sigma}(20d)$	3.84640×10^{-5}	4.89365×10^{-5}	0.7860
W/collimator-origin selected delayed rate	0	–	PASS